



ChipMixer CoinJoin-Style Case Study

A Bitcoin tracing case study showing how mixer/CoinJoin-style transactions weaken direct ownership tracing.

Case focus

This case starts from a ChipMixer-linked address discussed in research as appearing in thousands of CoinJoin transactions. The app searches that address transaction history and selects a transaction with strong CoinJoin-style structure.

The focus is not to prove the owner of every address. The focus is to show how many-input, many-output transactions with repeated output values can make simple tracing and common-input ownership assumptions unreliable.

Seed address: `bc1qs604c7jv6amk4cxqlnvuxv26hv3e48cds4m0ew`

Why this matters for tracing

In basic Bitcoin tracing, analysts may look at transaction inputs and outputs to infer who controlled which funds. Mixer and CoinJoin-style transactions are designed to disrupt that logic.

When a transaction combines many inputs and creates many similar outputs, it becomes harder to map one input to one output. It also weakens the common-input ownership heuristic, because the inputs may come from different participants rather than one wallet.

Transactions analysed	Transaction inputs	Transaction outputs	Repeated output groups	Total output BTC
150	137	312	11	245.5678

Key finding

The selected transaction contains **137 inputs** and **312 outputs**, showing pooled movement rather than a simple payment.

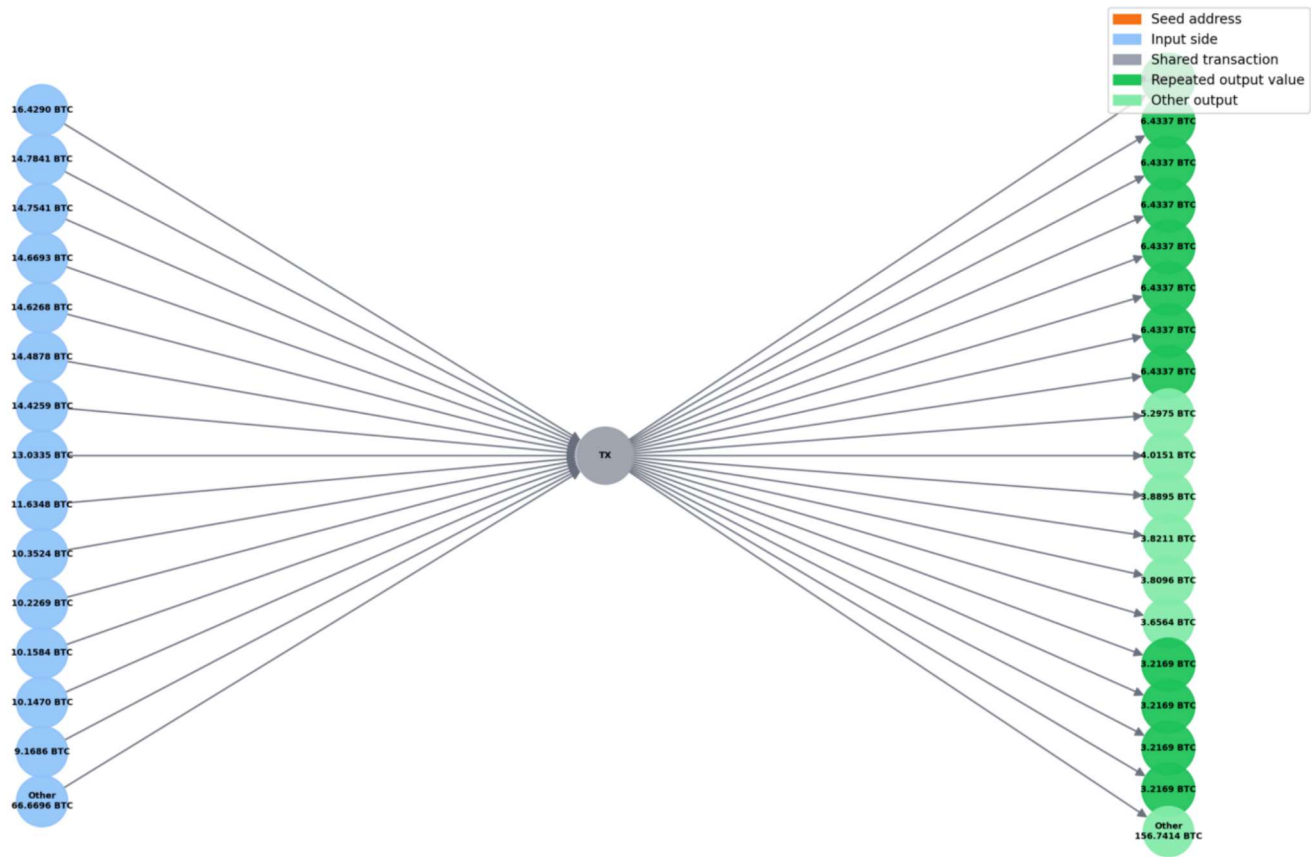
The strongest signal is the repeated output denominations. The app detected **11 repeated output value groups**, including repeated standard-sized outputs.

This matters because repeated output sizes make it harder to directly match one input participant to one output recipient. A participant may receive value back as several smaller outputs instead of one.

1. CoinJoin-style transaction graph

This graph shows the selected transaction structure. Green output nodes highlight repeated BTC denominations, which are important for understanding mixer-style redistribution.

CoinJoin-style transaction structure: many inputs, many outputs and repeated output values



2. Selected transaction summary

Metric	Value
Selected transaction	f398210da29d83cccd02651301156445c3173b2601801ec537bd
Date	03 Sep 2019
Input count	137
Output count	312
Total input BTC	245.56825319 BTC
Total output BTC	245.56783467 BTC
Transaction fee	0.00041852 BTC
Repeated output values detected	11
Largest repeated output group	98

3. Repeated output value summary (key evidence)

This table is the strongest evidence in this case. It shows repeated BTC denominations across many outputs, which is consistent with redistribution in standard-sized chunks.

Rounded Output BTC	Count	Total BTC
0.1005	98	
0.2011	35	
0.4021	26	
0.8042	21	
1.6084	20	
3.2169	19	
6.4337	8	
0.1979	3	
0.1003	2	
0.1002	2	

4. Address summary

Metric	Value
ChipMixer-linked seed address	bc1qs604c7jv6amk4cxqInvuxv26hv3e48cds4m0ew
Fetchd transactions analysed	150
Confirmed transaction count	5373
Confirmed BTC received	75.88853188 BTC
Confirmed BTC spent	75.88853188 BTC

▼ Show selected transaction inputs and outputs

Inputs

Input Index	Input Address	Input BTC	Seed Ad
0	bc1q8xuyyddm4pxd3vqkj5nkmnvcy5p6nmwc98haq6	0.3326	
1	bc1qp03chl6dxkqekt9ygctc99g9hvf8lypjm6876a	0.2001	
2	bc1q0w9vzqve7n8vr4yly6vck608xpgcfsxkpp5c00	10.1584	
3	bc1qq8fjxlvpv3ydlh5k4edr74lyffmz3673mj3az	0.0826	
4	bc1qh66sa6r868cpzdm3aj9atmeljld59xh8tnx8ed	0.0042	
5	bc1qxhnq07s8xut5ky5hxt5ndmxhhyg4lkkzsng0h	0.4022	
6	bc1qkkv4gmau6headg33fh9mqmkwe0eu2dht4uugsp	0.0079	
7	bc1q5gwn554gswfr937yxczyk83wsa9crqwrstf3t	0.102	
8	bc1qx8hx2ykm4ptgc7wcj7m8favjcndny3kgp27k2	0.0012	
9	bc1q8avp7lrs5s40je0numw2z7gxq8ynwsmdgx9kv0	16.429	

Outputs

Output Index	Output Address	Output BTC	Seed Addr
53	bc1qehshsm3hja40x3gs2pzzyjx9g6ddpzmk4hz4nn	0.1005	
54	bc1q53czpesqv2rnlr4mxzp20qevxxr2ndl5zl5aug	0.1005	
55	bc1qvd9fha9evs766hxx63gqsyvjswyzeyv8r4peku	0.1005	
56	bc1q6jnsu4tpxfk5830kt53aj3mdyfbz8z4w759mqsz	0.1005	
57	bc1q8hzz9wu57mgh0d8dqtsl4enp740pwgpay6vng8	0.1005	
58	bc1q279w3fnpr2p7g4lnvjspeqad9vrsrhp5es0qln	0.1005	
59	bc1q5anqsn2nu6ps8pmcsd7s6mmtgyq9luuxk5etdn	0.1005	
60	bc1qw654qdqpumrcq54ewfk0wgx7d2lp6007aacf35	0.1005	
61	bc1qqmcr0p2wmys5exwh3uus3zntgjdyk2rwqwr788	0.1005	
62	bc1qxwpw0c6tj52y455xdf2yuqd9xj55p3m3hpcjd9	0.1005	
63	bc1qd6hh08rc4a2vn9zmravwf5f99ae4vkn0lcr5vtn	0.1005	

Interpretation note

Blockchain tracing shows transaction movement, not real-world identity. In this case, the graph highlights structure: many inputs, many outputs and repeated output values. These are tracing complications rather than proof of who controlled each address.

References

▼ Show references and data sources

Blockstream (n.d.) *Blockstream API documentation*. Available at: <https://blockstream.info/explorer-api>.

U.S. Department of Justice (2023) *ChipMixer domain seized by law enforcement*. Available at: <https://www.justice.gov/opa/pr/chipmixer-domain-seized-law-enforcement>.

Europol (2023) *ChipMixer taken down for laundering criminal proceeds*. Available at: <https://www.europol.europa.eu/media-press/newsroom/news/multi-billion-euro-crypto-laundering-service-chipmixer-taken-down>.